

Ethical Machine Learning and Data Science

Assignment 1

Question 1

- a) In a few sentences, summarise why the sociologist might have chosen a randomised response design for this study. Your response should cover both the benefits and drawbacks of this choice. [3]

Randomised response designs are used when the information being collected from a study may be sensitive. By masking the identity of the respondents, through randomising the responses, the sociologist can preserve anonymity. With a respondents anonymity preserved, we find that individuals respond more truthfully, and refuse to respond to a question less, therefore reducing non-responsive answers and lying due to social desirability. The disadvantage to this method is that we introduce noise into the sample by randomising, therefore adding a larger variance, as well as the need for larger sample sizes to have an effective study.

- b) Using the data provided in criminality.csv, find the method of moments estimator for $p_r = \Pr(C = 1|H = 1)$. Hence provide an estimate and approximate 95% confidence interval for the probability that a right-handed individual has committed a crime, reporting your answers to two decimal places. [5]

We are given that $q = 0.2$. For this randomised response design, each participant will either respond with a fixed response with a probability $q = 0.2$, or else respond directly with a probability of $1 - q = 0.8$.

Therefore the reported response R satisfies:

$$R = \begin{cases} 1, & \text{with probability } q = 0.2, \\ C, & \text{with probability } 1 - q = 0.8, \end{cases} \quad (1)$$

where $C \in \{0, 1\}$ denotes true criminal activity.

Now we have:

$$\theta_r := \Pr(C = 1|H = 1) = q + (1 - q)\Pr(C = 1|H = 1) = q + (1 - q)p_r, \quad (2)$$

Where $p_r = \Pr(C = 1|H = 1)$. Now we solve for p_r , giving us:

$$p_r = \frac{\theta_r - q}{1 - q} \quad (3)$$

We can work out the method-of-moments estimator by equating θ_r for the sample mean of R among the right handed participants, which I calculated by using the following formula in a spreadsheet (LibreOffice Used):

$$= \text{AVERAGEIF}(A2 : A151, 1, B2 : B151) \quad (4)$$

This gives us:

$$\bar{R}_r = 0.462809917355372 \quad (5)$$

Now using:

$$\hat{p}_r = \frac{\bar{R}_r - q}{1 - q} \quad (6)$$

Where we equated θ_r to $\bar{R}_r$ to find the method-of-moments estimator, we find the estimator:

$$\hat{p}_r = 0.328512396694215 \approx 0.33(2dp) \quad (7)$$

Now to approximate the 95% confidence interval for we recognise that the design is a random variable that takes in values $\{0, 1\}$ with probability θ_r of 1 and $1 - \theta_r$ of 0, and so:

$$R|H = 1 \sim \text{Bernoulli}(\theta_r) \quad (8)$$

and so we have:

$$\text{Var}(\bar{R}_r) \approx \frac{\theta_r(1 - \theta_r)}{n_r} \quad (9)$$

We can find the standard error using:

$$\text{SE}(\hat{p}_r) \approx \frac{\sqrt{\bar{R}_r(1 - \bar{R}_r)/n_r}}{1 - q} \approx 0.0566 \quad (10)$$

and finally, we can calculate the confidence interval using the normal approximation, giving:

$$\hat{p}_r \pm 1.96\text{SE}(\hat{p}_r) = 0.3285 \pm 1.96(0.0566) \approx (0.22, 0.44) \quad (11)$$

So based on our available data, our method-of-moments estimator $\hat{p}_r = 0.33$ and the interval $(0.22, 0.44)$ covers the true value with approximately 95% confidence, all to two decimal places.

- c) Hence or otherwise, derive the method of moments estimator and construct a 95% confidence interval for $p = \Pr(C = 1)$ the overall prevalence of criminal activity. [5]

We use the same method as in b), but now applied to $\Pr(R = 1)$. First we find the sample mean $\bar{R}$, using a spreadsheet to compute this from our csv. Using LibreOffice:

$$\bar{R} = \text{AVERAGE}(B : B) = 0.453 \quad (12)$$

Recall from b)

$$\hat{p} = \frac{\bar{R} - q}{1 - q}, q = 0.2 \quad (13)$$

Using this, we calculate the method of moments estimator:

$$\hat{p} = 0.316 \approx 0.32(2dp) \quad (14)$$

Now we calculate the standard error:

$$\text{SE}(\hat{p}) = \frac{\sqrt{\bar{R}(q - \bar{R})/n}}{1 - q} = 0.0508082816229941 \approx 0.05(2dp) \quad (15)$$

Finally we calculate the confidence interval using the normal approximation:

$$\hat{p} \pm 1.96\text{SE}(\hat{p}) = 0.3167 \pm 1.96(0.0508) = (0.2171, 0.4163) \approx (0.22, 0.42) \quad (16)$$

So based on our available data, our method-of-moments estimator $\hat{p} = 0.32$ and the interval $(0.22, 0.42)$ covers the true value with approximately 95% confidence, all to two decimal places.

- d) Test at the 5% level whether there is a significant difference in the chance of criminal activity between left-handed and right-handed people. [4]

Let $p_h = \Pr(C = 1 \mid H = h)$ for $h \in \{0, 1\}$. We choose our hypothesis and test

$$H_0 : p_0 = p_1 \quad \text{vs} \quad H_1 : p_0 \neq p_1. \quad (17)$$

Under the forced-response design with $q = 0.2$,

$$\Pr(R = 1 \mid H = h) = q + (1 - q)p_h, \quad (18)$$

so the method-of-moments estimator is

$$\hat{p}_h = \frac{\bar{R}_h - q}{1 - q}, \quad (19)$$

where $\bar{R}_h$ is the sample mean of R within group $H = h$.

From b) and c) we have the data for H_1 , we calculate the relevant information for H_0 from our data:

$$n_1 = 121, \quad \bar{R}_1 = 0.4628, \quad n_0 = 29, \quad \bar{R}_0 = 0.4138. \quad (20)$$

and so:

$$\hat{p}_1 = \frac{0.4628 - 0.2}{0.8} = 0.3285, \quad \hat{p}_0 = \frac{0.4138 - 0.2}{0.8} = 0.2672. \quad (21)$$

Since $R \mid H = h \sim \text{Bernoulli}(\theta_h)$ with $\theta_h = q + (1 - q)p_h$, we have

$$\text{Var}(\hat{p}_h) \approx \frac{\bar{R}_h(1 - \bar{R}_h)}{n_h(1 - q)^2}. \quad (22)$$

Thus

$$\text{SE}(\hat{p}_1) \approx 0.0567, \quad \text{SE}(\hat{p}_0) \approx 0.1143. \quad (23)$$

Assuming independence between groups,

$$\text{SE}(\hat{p}_1 - \hat{p}_0) = \sqrt{\text{SE}(\hat{p}_1)^2 + \text{SE}(\hat{p}_0)^2} \approx 0.1276. \quad (24)$$

Using the Z-test (Wald test) statistic:

$$Z = \frac{\hat{p}_1 - \hat{p}_0}{\text{SE}(\hat{p}_1 - \hat{p}_0)} = \frac{0.3285 - 0.2672}{0.1276} \approx 0.48. \quad (25)$$

The two-sided p -value is

$$2\{1 - \Phi(|Z|)\} \approx 0.63. \quad (26)$$

Since $|Z| < 1.96$ (equivalently $p > 0.05$), we fail to reject H_0 at the 5% level. There is no statistically significant evidence of a difference in criminal activity prevalence between left-handed and right-handed individuals.

- e) Define what it means for a classifier to satisfy predictive parity, both in general and in this specific context. [2]

From the lecture notes:

Predictive parity ensures that the positive predictive value (precision) of the classifier is constant for all values of the fairness attribute.

$$\Pr(Y = 1 \mid \hat{Y} = 1, A = a_i) = \Pr(Y = 1 \mid \hat{Y} = 1, A = a_j) \quad \forall a_i, a_j \in \mathcal{A}. \quad (27)$$

In this specific context predictive parity requires:

$$\Pr(C = 1 \mid \hat{C} = 1, H = 0) = \Pr(C = 1 \mid \hat{C} = 1, H = 1) \quad (28)$$

Or among individuals that are predicted to have committed a crime, the true prevalence of criminal activity must be the same for both right-handed and left-handed individuals.

- f) Conduct an appropriate statistical assessment to determine whether the sociologist's classifier satisfies predictive parity. [6]

To have predictive parity for left-handedness and right-handedness we need:

$$\Pr(C = 1 \mid \hat{C} = 1, H = 0) = \Pr(C = 1 \mid \hat{C} = 1, H = 1). \quad (29)$$

Under the design of the experiment, we can formulate the following:

$$\Pr(R = 1 \mid \hat{C} = 1, H = h) = q + (1 - q) \Pr(C = 1 \mid \hat{C} = 1, H = h). \quad (30)$$

Define

$$\pi_h := \Pr(C = 1 \mid \hat{C} = 1, H = h), \quad \theta_h := \Pr(R = 1 \mid \hat{C} = 1, H = h). \quad (31)$$

Then

$$\pi_h = \frac{\theta_h - q}{1 - q}. \quad (32)$$

Estimate θ_h by the sample mean of R within the subgroup. Again we use a spreadsheet to run our calculations through LibreOffice. Equations used are trivial, and follow those used in b) and c). $\{H = h, \hat{C} = 1\}$:

$$\bar{R}_{h, \hat{C}=1} = \frac{1}{n_{h, \hat{C}=1}} \sum_{i: H_i=h, \hat{C}_i=1} R_i, \quad \hat{\pi}_h = \frac{\bar{R}_{h, \hat{C}=1} - q}{1 - q}. \quad (33)$$

From data/calculations we have:

$$n_{0, \hat{C}=1} = 7, \quad \bar{R}_{0, \hat{C}=1} = 0.4286, \quad n_{1, \hat{C}=1} = 28, \quad \bar{R}_{1, \hat{C}=1} = 0.5357. \quad (34)$$

and so we can calculate:

$$\hat{\pi}_0 = \frac{0.4286 - 0.2}{0.8} = 0.2857, \quad \hat{\pi}_1 = \frac{0.5357 - 0.2}{0.8} = 0.4196. \quad (35)$$

We choose our hypothesis, and test with a Z-test/Wald test, similar to part d):

$$H_0 : \pi_0 = \pi_1 \quad \text{vs} \quad H_1 : \pi_0 \neq \pi_1. \quad (36)$$

Since $R \mid (H = h, \hat{C} = 1)$ is Bernoulli with parameter θ_h ,

$$\text{SE}(\hat{\pi}_h) \approx \frac{\sqrt{\bar{R}_{h, \hat{C}=1}(1 - \bar{R}_{h, \hat{C}=1})/n_{h, \hat{C}=1}}}{1 - q}. \quad (37)$$

Thus

$$\text{SE}(\hat{\pi}_0) \approx 0.2338, \quad \text{SE}(\hat{\pi}_1) \approx 0.1178. \quad (38)$$

Assuming independence between handedness groups,

$$\text{SE}(\hat{\pi}_1 - \hat{\pi}_0) = \sqrt{\text{SE}(\hat{\pi}_1)^2 + \text{SE}(\hat{\pi}_0)^2} \approx 0.2618. \quad (39)$$

The statistic is:

$$Z = \frac{\hat{\pi}_1 - \hat{\pi}_0}{\text{SE}(\hat{\pi}_1 - \hat{\pi}_0)} = \frac{0.4196 - 0.2857}{0.2618} \approx 0.51. \quad (40)$$

The two-sided p -value is

$$2\{1 - \Phi(|Z|)\} \approx 0.61. \quad (41)$$

Since $|Z| < 1.96$ (equivalently $p > 0.05$), we fail to reject H_0 at the 5% level. There is no statistically significant evidence that the classifier violates predictive parity across handedness i.e there is no statistically significant evidence that among individuals that are predicted to have committed a crime, the true prevalence of criminal activity is different both right-handed and left-handed individuals..

Question 2

A national coffee shop chain is considering expansion and wants to determine the most suitable locations for new shops. Previously they have outsourced this to a consultant, who uses local demographic data and sales summaries from existing shops to assess the potential of new sites.

The coffee chain also operates a loyalty app, which records customers' order histories, dates of birth and postcodes. At present, this information is used solely for promotional offers but the company would like to provide this to the consultants so that it can be incorporated into the decision-making process for opening new shops.

In 1–2 pages (approximately 1500 words), outline the potential implications for customer privacy, and how these risks can be measured and mitigated. You should support your arguments with evidence from a range of relevant academic articles, case studies or professional guidelines. [10]

In this scenario, the consultant would be making a large move from storing lower identifiable data, consisting of aggregate area demographics and store sales, to a highly individualised dataset with sweeping risks across a number of fields, leading to a range of concerns. Introducing the new systems would lead to collection and storage of behavioural customer data through their transactional patterns: When they buy items, what they buy, how much they spend and the implications of these transactions. As well as this, the new system would collect unique identifiable information about the customer; their DOB and postcode. This furthers the privacy risks of the customer. The personal data collected here is both a significant legal risk under UK GDPR as well as a harm risk, one which can lead to intrusion on the customers privacy, lead to discrimination by statistics, and leave the coffee chain open to risks involving the loss of control over data.[7] This results in a large shift from aggregate market analysis, into tailored and individualised market analytics, and with it, a whole host of different compliance and harmful risks.

As this new data is introduced and collected, important changes happen to the data framework of the company, which must be accounted for. Firstly, the introduction of granular data allowing for behavioural traces, that of collecting customer spending patterns, poses unique risks to identifying individuals. Even from small sets of data points, customers may be uniquely identified.[3] This is compounded by the second collection method, that of traits that represent unique demographic identifiers: postcodes, date of birth, sex. Studies have shown that even seemingly simple characteristics such as these pose a risk to uniquely identify individuals.[12] These characteristics, patterns and traces lead to direct and indirect identifiers for customers, a large change in comparison to the previous data collection system. The consultants may further choose to combine these data with publicly available datasets, enabling further identification, but also the threat of malignant inference. As information linkage increases, so does the risk of identifying sensitive customer information, regardless of anonymisation techniques.[11] Similarly, this increase in available information, when applied with analytical algorithms and machine learning can lead to harmful prejudicial and systematic bias towards customers, and leading to a pattern of discrimination that may alienate customers as well as pose legal risk, a risk that was significant when analysing facial recognition software.[1]

Expanding on these risks, we break down the potential implications into five categories, Identification Risk, Inference Risk, Structural Inequalities, Cultural and Societal implications, and Regulatory and Legal Risk.

First, identification risk. Privacy is an ever present topic in the information age, and research has shown that customers can be resistant towards engaging with a business if privacy trust is perceived to be violated.[2] Moving to a highly data collection dependent system may pose significant business risks from multiple faces. Customers may choose to engage with other businesses, particularly if data engagement is hard to opt-out of. Incorrect handling of anonymity can cause reputational backlash and risk, regulatory and legal risks and infringement, and even invite malignant targeting from entities, in the form of targeted hacking and data breaches. Whether by intent, the ability to identify customers from a dataset poses significant implications. While the intention of this data collection may be to perform market analytics, enhancing customer experience and providing insight towards key growth analytics through inference, this does not come without implications. Function creep can lead to businesses prioritising harmful strategies that violates a users trust, and invites legal and reputational risk, such as selling

data to third-parties for inference.[8] Even if a business does not decide to sell on your data, over-eager analytics can invite discriminatory pricing and targeting through unchecked inferences.[1] Unchecked inferences can lead to major business implications, in the form of structural inequalities. As shown in [1], biased data distributions can lead to under-representation of specific minority groups. This can further lead to missing growth opportunities in communities represented by these groups, contributing to exclusion of these communities. The societal structural inequalities posed here are large, not only an exclusion of minorities from becoming customers, but impactful loss of jobs and opportunities within these communities. This ties in directly with collective and societal implications. Particularly with large businesses, the exclusion of under-represented demographics may lead to collective geographic profile reshaping. A coffee shop is often seen as a third space, and these growth factors may lead to less connections between groups, contributing to isolation of urban and economically underprivileged groups. This becomes a self-fulfilling prophecy, further denying opportunities for these groups, through network and connections.[10] Finally, more directly, moving to this system poses huge regulatory and legal risk. Even with mitigations, pseudonymity and de-anonymising methods do not exclude businesses from GDPR obligations[7], and failure to comply by these regulations invites legal risk, reputational risk, as well as regulatory fines and oversight, all of which could severely hinder the growth of a business, costing people jobs, community and opportunities.[5]

With all of the risks thoroughly explored, we now pose the question: How do we effectively measure these risks? Thorough operational methodology must be explored and implemented to allow the consultants to find exposed risks, and mitigate these effectively.

For identification risk, it is crucial for a business and their consultants to be able to assess if a customer can be identified from their dataset. The implications are clear - if they are unable to be identified, then a mitigation is not required, but more than likely - heavy mitigation methods are required. This can be assessed using a k-anonymity framework[13], treating postcode, sex and date of birth as semi-identifiers, and reporting the population of records that are unique or near unique. Similarly, [11] shows that behavioural patterns with auxiliary knowledge can also identify customers. A risk assessment framework can be useful when measuring risk in this scenario, attempting to record linkage between the business' dataset and public datasets, and measuring match confidence.[4][11]

Measuring inference risk is harder, but [13] provides methods for preventing attribute inference through limits of k-anonymity. Similarly, [9] provides a novel method for reducing closeness between identifying attributes, reducing the impact of malignant inference. The underlying risk is clear, even when identities are protected, high homogeneity within equivalence classes can increase this inference risk. Testing predictive accuracy can also highlight this risk, with [7] and [11] highlighting that identifiability includes inference risk, not just exposing and identifying individuals, but deducing behaviours that may pose harmful risks.

Consultants can also choose to measure bias and representation risk, help mitigate potential structural inequalities, and cultural and societal implications. The aim would be to measure error rates across demographics, as well as try to identify if neighbourhoods and demographics are systematically under-represented and predicted.[1] provides some insight into potential analysis of the sort. Finally, for Regulatory and Legal risk, a suitable risk assessment and framework would be set up to map if the collection of data, its underlying nature, processing and third-party sharing are in line with current regulations. [7] and similar ICO guidelines present detail regulatory requirements in the UK.

Following the assessment of these risks, proper procedural mitigations would have to be applied to ensure that these risks are accounted for, and proper oversight is applied in a resilient framework. First, the business would ensure that unnecessary data is removed, in order to comply with GDPR guides on data minimisation.[6] Similarly, data protection guidelines must be followed and appropriate systems put in place, to ensure data handling is robust. Once legal and data frameworks are in place, the business would need to mitigate identification and inference risks through statistical methods, aiming to avoid attribute disclosures through distributional constraints. This would mean that the datasets limit the amount of identification and inference, through anonymisation and similar methods. Again, frameworks and methods are shown in [11][13][9].

To conclude, even if commercial viability can be proven to be overwhelmingly positive, a move to a new system incorporating data into the decision-making process can have tremendous impacts. Risks

can be amplified, and if not assessed and mitigated correctly, they can be crippling for a business. The decision to incorporate a robust data collection and market inference framework is not based on just commercial interest, but must demonstrate necessity, risk reduction, accountability and adherence to regulatory and risk frameworks.

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